

OUTPUT SCREENSHOTS

SARATHY P

QUERY 1 : What are the pricing plans available for your software?

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PROBLEMS  PORTS
• PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& c:\Users\Madava\Downlo
te.ps1)
• (venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
[Memory] SQLite database ready at: C:\Users\Madava\Downloads\IBM_SARATHY_DA-2\memory.db

[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.

=====

ABC TECHNOLOGIES – AI CUSTOMER SUPPORT SYSTEM
Built with LangGraph

=====

Please enter your details to begin.

Your name      : Sarathy
Your customer ID: 1

Welcome, Sarathy! (ID: 1)
Type your support query below. Type 'quit' to exit.

-----

Your query : What are the pricing plans available for your software

[System] Processing your query...

[Intent] Classifying query: 'what are the pricing plans available for your software'
[Memory] Retrieved 0 history entries for '1'.
[Intent] Classified as: Sales
[Router] Intent='Sales' → routing to 'sales_agent'
[Sales Agent] Processing query...
[RAG] Loaded 'company_policy' as 7 chunk(s).
[RAG] Loaded 'pricing_guide' as 8 chunk(s).
[RAG] Loaded 'technical_manual' as 14 chunk(s).
[RAG] Loaded 'faq' as 22 chunk(s).
[RAG] Loaded 4 knowledge-base documents -> 51 total chunks.
```

```
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
[kept] score=2 source=pricing_guide
[kept] score=2 source=technical_manual

[Supervisor] Reviewing draft response...
[Supervisor] Draft passes all quality checks ✓
[Supervisor] Notes: Draft approved without modifications.
[Memory] Saved interaction for customer '1'.
[Supervisor] Interaction saved to memory for customer '1'.

=====

Customer   : Sarathy (1)
Query      : What are the pricing plans available for your software
Intent     : Sales
Department : Sales
Supervisor : Draft approved without modifications.

-----
RESPONSE:
-----

Hello Sarathy,

Thank you for contacting ABC Technologies Sales.

I understand your query: "What are the pricing plans available for your software"

Based on our knowledge base, here is the information relevant to your request:

[Source: pricing_guide]
ABC TECHNOLOGIES - PRICING GUIDE
=====
1. SUBSCRIPTION PLANS
-----
STARTER PLAN - $29/month (or $290/year - save 17%)
- Up to 5 users
- 10 GB cloud storage
- Basic reporting
- Email support
- Core business management features
- Mobile app access
```

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PROBLEMS  PORTS
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py

---

[Source: pricing_guide]
3. FREE TRIAL
-----
- 14-day free trial available for all plans
- No credit card required
- Full feature access during trial
- Data retained after trial ends if upgraded

---

[Source: technical_manual]
8. BACKUP & DATA RECOVERY
-----
- Automatic daily backups for all plans
- Manual backup export: Settings > Data > Export Backup
- Recovery window: 30 days for Starter, 90 days for Pro, 365 days for Enterprise
- Point-in-time recovery available for Enterprise plan

Please do not hesitate to reach out if you need further assistance.

Best regards,
Alex | Sales Specialist
ABC Technologies Sales

=====

Do you have another query? (yes/no) : no

=====

Thank you for contacting ABC Technologies, Sarathy!
Total queries handled this session: 1
Your conversation history is saved under ID: 1

=====
```

QUERY -2 : I forgot my account password.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PROBLEMS  PORTS
• (venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
[Memory] SQLite database ready at: C:\Users\Madava\Downloads\IBM_SARATHY_DA-2\memory.db

[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.

=====

ABC TECHNOLOGIES – AI CUSTOMER SUPPORT SYSTEM
Built with LangGraph

=====

Please enter your details to begin.

Your name      : Sarathy
Your customer ID: 1

Welcome, Sarathy! (ID: 1)
Type your support query below. Type 'quit' to exit.

-----
Your query : I forgot my account password

[System] Processing your query...

[Intent] Classifying query: 'I forgot my account password'
[Memory] Retrieved 2 history entries for '1'.
[Intent] Classified as: Account
[Router] Intent='Account' → routing to 'account_agent'
[Account Agent] Processing query...
[RAG] Loaded 'company_policy' as 7 chunk(s).
[RAG] Loaded 'pricing_guide' as 8 chunk(s).
[RAG] Loaded 'technical_manual' as 14 chunk(s).
[RAG] Loaded 'faq' as 22 chunk(s).
[RAG] Loaded 4 knowledge-base documents -> 51 total chunks.
[RAG] Retrieved 3 chunk(s) for query: 'I forgot my account password'
[kept] score=3 source=technical_manual
```

```
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
[RAG] Loaded 'faq' as 22 chunk(s).
[RAG] Loaded 4 knowledge-base documents -> 51 total chunks.
[RAG] Retrieved 3 chunk(s) for query: 'I forgot my account password'
      [kept] score=3 source=technical_manual
      [kept] score=3 source=faq
      [kept] score=3 source=faq

[Supervisor] Reviewing draft response...
[Supervisor] Draft passes all quality checks ✓
[Supervisor] Notes: Draft approved without modifications.
[Memory] Saved interaction for customer '1'.
[Supervisor] Interaction saved to memory for customer '1'.
```

```
=====
Customer   : Sarathy (1)
Query      : I forgot my account password
Intent     : Account
Department : Account Management
Supervisor : Draft approved without modifications.
```

```
-----
RESPONSE:
-----
```

Hello Sarathy,

Thank you for contacting ABC Technologies Account Management.

I understand your query: "I forgot my account password"

Based on our knowledge base, here is the information relevant to your request:

[Source: technical_manual]

3. LOGIN ISSUES

-
- Forgotten Password: Click "Forgot Password" on the login page; a reset link is sent to registered email
 - Account Locked: Accounts are locked after 5 failed login attempts; contact support or wait 30 minutes
 - Two-Factor Authentication (2FA): Use the Authenticator app or SMS code sent to registered phone
 - SSO Login Issues: Ensure your organization's SSO provider is configured correctly in Admin Settings

OUTPUT DEBUG CONSOLE TERMINAL PROBLEMS PORTS

```
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
```

[Source: faq]
ACCOUNT QUESTIONS

Q: How do I reset my password?

A: Click "Forgot Password" on the login page. A reset link will be emailed to your registered address. The link expires in 24 hours.

[Source: faq]

Q: Why can't I log in?

A: Ensure you're using the correct email and password. After 5 failed attempts, the account is locked for 30 minutes. Use "Forgot Password" if needed.

Please do not hesitate to reach out if you need further assistance.

Best regards,
Riley | Account Support Specialist
ABC Technologies Account Management

```
=====
Do you have another query? (yes/no) : no
=====
```

Thank you for contacting ABC Technologies, Sarathy!
Total queries handled this session: 1
Your conversation history is saved under ID: 1

QUERY -3 : My application crashes whenever I upload a file.

```
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
[Memory] SQLite database ready at: C:\Users\Madava\Downloads\IBM_SARATHY_DA-2\memory.db
```

```
[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.
```

```
=====

ABC TECHNOLOGIES – AI CUSTOMER SUPPORT SYSTEM
Built with LangGraph

=====
```

Please enter your details to begin.

Your name : Lathika
Your customer ID: 2

Welcome, Lathika! (ID: 2)
Type your support query below. Type 'quit' to exit.

Your query : My application crashes whenever I upload a file

[System] Processing your query...

```
[Intent] Classifying query: 'My application crashes whenever I upload a file'
[Memory] Retrieved 0 history entries for '2'.
[Intent] Classified as: Technical
[Router] Intent='Technical' -> routing to 'technical_agent'
[Technical Agent] Processing query...
[RAG] Loaded 'company_policy' as 7 chunk(s).
[RAG] Loaded 'pricing_guide' as 8 chunk(s).
[RAG] Loaded 'technical_manual' as 14 chunk(s).
[RAG] Loaded 'faq' as 22 chunk(s).
[RAG] Loaded 4 knowledge-base documents -> 51 total chunks.
[RAG] Retrieved 3 chunk(s) for query: 'My application crashes whenever I upload a file'
[kept] score=3 source=technical_manual
```

OUTPUT DEBUG CONSOLE TERMINAL PROBLEMS PORTS

```
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
[kept] score=3 source=technical_manual
[kept] score=2 source=faq
[kept] score=2 source=faq
[RAG] Retrieved 2 chunk(s) for query: 'file upload troubleshooting application error'
[kept] score=3 source=technical_manual
[kept] score=2 source=technical_manual
```

```
[Supervisor] Reviewing draft response...
[Supervisor] Draft passes all quality checks ✓
[Supervisor] Notes: Draft approved without modifications.
[Memory] Saved interaction for customer '2'.
[Supervisor] Interaction saved to memory for customer '2'.
```

```
=====

Customer : Lathika (2)
Query : My application crashes whenever I upload a file
Intent : Technical
Department : Technical Support
Supervisor : Draft approved without modifications.

=====
```

RESPONSE:

Hello Lathika,

Thank you for contacting ABC Technologies Technical Support.

I understand your query: "My application crashes whenever I upload a file"

Based on our knowledge base, here is the information relevant to your request:

```
[Source: technical_manual]
4. FILE UPLOAD TROUBLESHOOTING
-----
Common causes of file upload crashes:
- File size exceeds the plan limit (Starter: 25MB, Pro: 100MB, Enterprise: 500MB)
- Unsupported file format (supported: PDF, DOCX, XLSX, PNG, JPG, CSV, ZIP)
- Slow internet connection causing timeout
- Browser cache/cookies issue
```

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PROBLEMS  PORTS
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
Common causes of file upload crashes:
- File size exceeds the plan limit (Starter: 25MB, Pro: 100MB, Enterprise: 500MB)
- Unsupported file format (supported: PDF, DOCX, XLSX, PNG, JPG, CSV, ZIP)
- Slow internet connection causing timeout
- Browser cache/cookies issue

---

[Source: faq]
TECHNICAL QUESTIONS
-----
Q: Why is my application crashing?
A: Common causes include browser cache issues, unsupported file formats, or files exceeding size limits. Try clearing your cache or using a different browser.

---

[Source: faq]
Q: What file formats are supported for upload?
A: PDF, DOCX, XLSX, PNG, JPG, JPEG, CSV, and ZIP files are supported.

--- Additional Troubleshooting ---
[Source: technical manual]
5. APPLICATION ERRORS
-----
Error Code 500 (Internal Server Error):
- Refresh the page; if persists, clear cache
- Check status.abctech.com for outage notifications

Please do not hesitate to reach out if you need further assistance.

Best regards,
Sam | Technical Support Engineer
ABC Technologies Technical Support

=====

Do you have another query? (yes/no) : no

=====
```

QUERY -4: I need a refund for my annual subscription.

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PROBLEMS  PORTS
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
[Memory] SQLite database ready at: C:\Users\Madava\Downloads\IBM_SARATHY_DA-2\memory.db

[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.

=====

ABC TECHNOLOGIES – AI CUSTOMER SUPPORT SYSTEM
Built with LangGraph

=====

Please enter your details to begin.

Your name      : Nandana
Your customer ID: 3

Welcome, Nandana! (ID: 3)
Type your support query below. Type 'quit' to exit.

-----
Your query :   I need a refund for my annual subscription

[System] Processing your query...

[Intent] Classifying query: 'I need a refund for my annual subscription'
[Memory] Retrieved 0 history entries for '3'.
[Intent] Classified as: Billing
[Router] Intent='Billing' → routing to 'billing_agent'
[Billing Agent] Processing query...
[RAG] Loaded 'company_policy' as 7 chunk(s).
[RAG] Loaded 'pricing_guide' as 8 chunk(s).
[RAG] Loaded 'technical_manual' as 14 chunk(s).
[RAG] Loaded 'faq' as 22 chunk(s).
[RAG] Loaded 4 knowledge-base documents -> 51 total chunks.
[RAG] Retrieved 3 chunk(s) for query: 'I need a refund for my annual subscription'
[kept] score=3 source=company policy
```

```
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
[Router] High-risk request detected -> human approval required.

=====
● HUMAN-IN-THE-LOOP APPROVAL REQUIRED
=====
Customer ID : 3
Customer Name: Nandana
Intent : Billing
Department : Billing

Customer Query:
  I need a refund for my annual subscription

Draft Response (for supervisor review):
-----
Hello Nandana,

Thank you for contacting ABC Technologies Billing Support.

I have received your request regarding: "I need a refund for my annual subscription"

This request requires review and approval from our billing supervisor before it can be processed. Your case has been escalated and you will receive a response within 24 business days.

Reference information from our policy:

[Source: company_policy]
ABC TECHNOLOGIES - COMPANY POLICY DOCUMENT
=====
1. REFUND POLICY
-----
- Customers are eligible for a full refund within 30 days of purchase.
- Refunds after 30 days are evaluated on a case-by-case basis.
- Refund requests must be submitted through the support portal.
- Approved refunds are processed within 7-10 business days.
- Annual subscription refunds are prorated after the 30-day window.

---

[Source: company_policy]
```

Human Approval : Accepted

```
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
Reference information from our policy:

[Source: company_policy]
ABC TECHNOLOGIES - COMPANY POLICY DOCUMENT
=====
1. REFUND POLICY
-----
- Customers are eligible for a full refund within 30 days of purchase.
- Refunds after 30 days are evaluated on a case-by-case basis.
- Refund requests must be submitted through the support portal.
- Approved refunds are processed within 7-10 business days.
- Annual subscription refunds are prorated after the 30-day window.

---

[Source: company_policy]
2. SUBSCRIPTION CANCELLATION POLICY
-----
- Customers may cancel their subscription at any time.
- Cancellation takes effect at the end of the current billing cycle.
- No partial refunds for mid-cycle cancellations unless under 30-day policy.
- Cancellation requests must be confirmed via email.

---

[Source: company_policy]
4. COMPENSATION POLICY
-----
- Compensation may be offered for verified service outages exceeding 4 hours.
- Compensation is in the form of service credits, not cash.
- Compensation requests must include outage reference number.
- Maximum compensation is one month of subscription fee.

Best regards,
Jordan | Billing Support Specialist
ABC Technologies Billing
-----

Supervisor Decision - type 'approve' or 'reject': approve
```

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PROBLEMS  PORTS
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
[Human Approval] Decision recorded: APPROVED
=====

[Router] Request approved by supervisor -> continuing to response.

[Supervisor] Reviewing draft response...
[Supervisor] Draft passes all quality checks ✓
[Supervisor] Notes: Draft approved without modifications.
[Memory] Saved interaction for customer '3'.
[Supervisor] Interaction saved to memory for customer '3'.
=====

Customer   : Nandana (3)
Query      : I need a refund for my annual subscription
Intent     : Billing
Department : Billing
Approval   : APPROVED
Supervisor : Draft approved without modifications.
-----
RESPONSE:
-----
Hello Nandana,

Thank you for contacting ABC Technologies Billing Support.

I have received your request regarding: "I need a refund for my annual subscription"

This request requires review and approval from our billing supervisor before it can be processed. Your case has been escalated and you will receive a response within 24 business hours.

Reference information from our policy:

[Source: company_policy]
ABC TECHNOLOGIES - COMPANY POLICY DOCUMENT
=====
1. REFUND POLICY
-----
- Customers are eligible for a full refund within 30 days of purchase.
- Refunds after 30 days are evaluated on a case-by-case basis.
```

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PROBLEMS  PORTS
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
- Customers are eligible for a full refund within 30 days of purchase.
- Refunds after 30 days are evaluated on a case-by-case basis.
- Refund requests must be submitted through the support portal.
- Approved refunds are processed within 7-10 business days.
- Annual subscription refunds are prorated after the 30-day window.

---

[Source: company_policy]
2. SUBSCRIPTION CANCELLATION POLICY
-----
- Customers may cancel their subscription at any time.
- Cancellation takes effect at the end of the current billing cycle.
- No partial refunds for mid-cycle cancellations unless under 30-day policy.
- Cancellation requests must be confirmed via email.

---

[Source: company_policy]
4. COMPENSATION POLICY
-----
- Compensation may be offered for verified service outages exceeding 4 hours.
- Compensation is in the form of service credits, not cash.
- Compensation requests must include outage reference number.
- Maximum compensation is one month of subscription fee.

Best regards,
Jordan | Billing Support Specialist
ABC Technologies Billing

=====

Do you have another query? (yes/no) : no

=====

Thank you for contacting ABC Technologies, Nandana!
Total queries handled this session: 1
Your conversation history is saved under ID: 3
```

Human Approval: **Rejected**

```
• (venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
[Memory] SQLite database ready at: C:\Users\Madava\Downloads\IBM_SARATHY_DA-2\memory.db
```

```
[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.
```

```
=====

ABC TECHNOLOGIES – AI CUSTOMER SUPPORT SYSTEM
Built with LangGraph

=====
```

Please enter your details to begin.

Your name : Pavindhar
Your customer ID: 4

Welcome, Pavindhar! (ID: 4)
Type your support query below. Type 'quit' to exit.

```
-----
Your query : I need a refund for my annual subscription
```

[System] Processing your query...

```
[Intent] Classifying query: 'I need a refund for my annual subscription'
[Memory] Retrieved 0 history entries for '4'.
[Intent] Classified as: Billing
[Router] Intent='Billing' → routing to 'billing_agent'
[Billing Agent] Processing query...
[RAG] Loaded 'company_policy' as 7 chunk(s).
[RAG] Loaded 'pricing_guide' as 8 chunk(s).
[RAG] Loaded 'technical_manual' as 14 chunk(s).
[RAG] Loaded 'faq' as 22 chunk(s).
[RAG] Loaded 4 knowledge-base documents -> 51 total chunks.
[RAG] Retrieved 3 chunk(s) for query: 'I need a refund for my annual subscription'
[kept] score=3 source=company_policy
```

```
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
[Router] High-risk request detected → human approval required.
```

```
=====
● HUMAN-IN-THE-LOOP APPROVAL REQUIRED
=====
```

Customer ID : 4
Customer Name: Pavindhar
Intent : Billing
Department : Billing

Customer Query:
I need a refund for my annual subscription

Draft Response (for supervisor review):

Hello Pavindhar,

Thank you for contacting ABC Technologies Billing Support.

I have received your request regarding: "I need a refund for my annual subscription"

This request requires review and approval from our billing supervisor before it can be processed. Your case has been escalated and you will receive a response within 24 business hours.

Reference information from our policy:

```
[Source: company_policy]
ABC TECHNOLOGIES - COMPANY POLICY DOCUMENT
=====
```

1. REFUND POLICY

- Customers are eligible for a full refund within 30 days of purchase.
- Refunds after 30 days are evaluated on a case-by-case basis.
- Refund requests must be submitted through the support portal.
- Approved refunds are processed within 7-10 business days.
- Annual subscription refunds are prorated after the 30-day window.

```
---
[Source: company_policy]
```



```
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
Reference information from our policy:

[Source: company_policy]
ABC TECHNOLOGIES - COMPANY POLICY DOCUMENT
=====
1. REFUND POLICY
-----
- Customers are eligible for a full refund within 30 days of purchase.
- Refunds after 30 days are evaluated on a case-by-case basis.
- Refund requests must be submitted through the support portal.
- Approved refunds are processed within 7-10 business days.
- Annual subscription refunds are prorated after the 30-day window.

---

[Source: company_policy]
2. SUBSCRIPTION CANCELLATION POLICY
-----
- Customers may cancel their subscription at any time.
- Cancellation takes effect at the end of the current billing cycle.
- No partial refunds for mid-cycle cancellations unless under 30-day policy.
- Cancellation requests must be confirmed via email.

---

[Source: company_policy]
4. COMPENSATION POLICY
-----
- Compensation may be offered for verified service outages exceeding 4 hours.
- Compensation is in the form of service credits, not cash.
- Compensation requests must include outage reference number.
- Maximum compensation is one month of subscription fee.

Best regards,
Jordan | Billing Support Specialist
ABC Technologies Billing
=====

Supervisor Decision -- type 'approve' or 'reject': reject
```

```
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py

[Human Approval] Decision recorded: REJECTED
=====

[Router] Request rejected by supervisor -> ending workflow.
[End] Request was rejected by supervisor. Sending rejection response.
[Memory] Saved interaction for customer '4'.

=====

Customer : Pavindhar (4)
Query : I need a refund for my annual subscription
Intent : Billing
Department : Billing
Approval : REJECTED
Supervisor : Request rejected by human supervisor.
=====

RESPONSE:
=====

Hello Pavindhar,

Thank you for contacting ABC Technologies Support.

After careful review, we are unable to process your request at this time. A support specialist will follow up with you within 24 business hours to discuss your options.

We apologize for any inconvenience and appreciate your patience.

Best regards,
ABC Technologies Support Team
=====

Do you have another query? (yes/no) : no
=====

Thank you for contacting ABC Technologies, Pavindhar!
Total queries handled this session: 1
Your conversation history is saved under ID: 4
```

QUERY -5: What was my previous support issue?

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PROBLEMS  PORTS
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
[Memory] SQLite database ready at: C:\Users\Madava\Downloads\IBM_SARATHY_DA-2\memory.db

[System] Starting AI Customer Support System...
[Graph] LangGraph workflow compiled successfully.
[System] System ready.

=====

ABC TECHNOLOGIES — AI CUSTOMER SUPPORT SYSTEM
Built with LangGraph

=====

Please enter your details to begin.

Your name      : Lathika
Your customer ID: 2

Welcome, Lathika! (ID: 2)
Type your support query below. Type 'quit' to exit.

-----

Your query : What was my previous support issue

[System] Processing your query...

[Intent] Classifying query: 'What was my previous support issue'
[Memory] Retrieved 2 history entries for '2'.
[Intent] Classified as: Memory
[Router] Intent='Memory' → routing to 'memory_recall_agent'
[Memory Recall Agent] Retrieving conversation history...

[Supervisor] Reviewing draft response...
[Supervisor] Draft passes all quality checks ✓
[Supervisor] Notes: Draft approved without modifications.
[Memory] Saved interaction for customer '2'.
[Supervisor] Interaction saved to memory for customer '2'.
```

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PROBLEMS  PORTS
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py

=====

Customer   : Lathika (2)
Query      : What was my previous support issue
Intent     : Memory
Department : Memory Recall
Supervisor : Draft approved without modifications.

-----

RESPONSE:
-----

Hello Lathika,

I've retrieved your previous support history from our records.

Your most recent support query was:
"My application crashes whenever I upload a file"

Full conversation history:

=== Previous Conversation History ===

[2026-06-28T14:38:42] [Technical Support]
Customer: My application crashes whenever I upload a file

[2026-06-28T14:38:42] [Technical Support]
Agent: Hello Lathika,

Thank you for contacting ABC Technologies Technical Support.

I understand your query: "My application crashes whenever I upload a file"

Based on our knowledge base, here is the information relevant to your request:

[Source: technical_manual]
4. FILE UPLOAD TROUBLESHOOTING
-----

Common causes of file upload crashes:
- File size exceeds the plan limit (Starter: 25MB, Pro: 100MB, Enterprise: 500MB)
- Unsupported file format (supported: PDF, DOCX, XLSX, PNG, JPG, CSV, ZIP)
- Slow internet connection causing timeout
```

```
(venv) PS C:\Users\Madava\Downloads\IBM_SARATHY_DA-2> python demo.py
```

```
[Source: faq]  
TECHNICAL QUESTIONS  
-----
```

Q: Why is my application crashing?

A: Common causes include browser cache issues, unsupported file formats, or files exceeding size limits. Try clearing your cache or using a different browser.

```
[Source: faq]
```

Q: What file formats are supported for upload?

A: PDF, DOCX, XLSX, PNG, JPG, JPEG, CSV, and ZIP files are supported.

--- Additional Troubleshooting ---

```
[Source: technical_manual]
```

5. APPLICATION ERRORS

Error Code 500 (Internal Server Error):

- Refresh the page; if persists, clear cache
- Check status.abctech.com for outage notifications

Please do not hesitate to reach out if you need further assistance.

Best regards,
Sam | Technical Support Engineer
ABC Technologies Technical Support

=== End of History ===

Is there anything else I can help you with today?

Best regards,
ABC Technologies Support Team

=====

Do you have another query? (yes/no) : no

^^